Dharmik Patel

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Education

University of Ottawa — Ottawa, Canada
Master of Science, Physics — Sep 2025 - Current
Advisor: Prof. Khabat Heshami

University of Toronto, St. George (University College) — Toronto, Canada
Honours Bachelor of Science, Mathematics and Physics — Sep 2020 - Jun 2024

Awards

University of Toronto International Scholar Award — University of Toronto
A merit-based scholarship valued at $100,000 CAD over four years.

Research Experience

Photonic Quantum Information Research — Ottawa, Canada
National Research Council of Canada & University of Ottawa — Aug 2025 -
Supervisors: Prof. Khabat Heshami & Dr. Milica Banić

- Conducting research on non-Gaussian resources as a member of the Photonic Quantum Information Processing (PQuIP) group at the National Research Council of Canada (NRC).

Gaussian/Non-Gaussian States: Generation and Applications — Toronto, Canada
University of Toronto — Jun 2024 - Aug 2025
Supervisors: Prof. John Sipe & Dr. Colin Vendromin

- Constructing non-Gaussian states via Gaussian resource states and extending previous results to generate non-Gaussian cluster states for photonic quantum computation. Studying the properties of rotationally symmetric Wigner-negative states to determine possible quantum error correcting codes.
- Simulating lossy nonlinear generation of multimode Gaussian states in systems with linear optics circuits and threshold detectors. Helped construct novel methods to calculate coincidence click probabilities in such systems using operator disentangling methods.

Undergraduate Thesis Research — Toronto, Canada
University of Toronto — Aug 2023 - Jun 2024
Supervisors: Prof. John Sipe & Dr. Colin Vendromin
Title: *A Unique Representation for the Unitary Time Evolution Operator of an N-Mode Quadratic Hamiltonian*
Wrote an exhaustive reconstruction of an existence and uniqueness proof for the titular representation as an operator product, with a focus on the mathematical foundations using Lie algebraic and operator calculus methods. As an application, wrote a novel Julia simulation of nonlinear generation of squeezed light in a coupled channel waveguide-ring resonator system via an effective $\chi^{(2)}$ process.

VLT Spectroscopy of Ultra-Faint Dwarf Galaxies — Toronto, Canada
University of Toronto — Apr 2022 - Aug 2022
Supervisor: Prof. Ting Li

- Wrote data classification algorithms using the `astropy` Python package to perform consistent reductions and measurements for three ultra-faint dwarf galaxies using archival data from the GIRAFFE spectrograph on the Very Large Telescope (VLT).

- Customized and optimized cross-matching algorithms using `astropy` to work in conjunction with the data classification algorithms, in close collaboration with the Near-Field Cosmology research group.

Teaching Experience

PHY1124 Laboratory TA Ottawa, Canada
University of Ottawa Jan 2026 - Current

- Leading lab sessions for first year engineering students. Duties include pre-lab preparation, in-lab experiment supervision and demonstration, and post-lab grading and feedback.

PHY1321/1321 First Year Laboratory TA Ottawa, Canada
University of Ottawa Sep 2025 - Dec 2025

- Leading lab sessions for first year physics students taking PHY1321/1321. Duties include pre-lab preparation, in-lab experiment supervision and demonstration, and post-lab grading and feedback.

Conferences and Talks

Q-SITE 2024 Toronto, Canada
Poster Presentation Sep 2024

- Presented a poster titled *Multimode Squeezed States in Silicon Nitride Ring Resonators* on lossy generation of multimode squeezed states via dual pump spontaneous four wave mixing (SFWM) processes.

uOttawa - NRC PQuIP Group Seminar Toronto, Canada
Seminar Talk Sep 2024

- Gave a talk to the University of Ottawa - National Research Council (NRC) Photonic Quantum Information Processing (PQuIP) research group, titled *Nonlinear Generation of Gaussian States and Applications* (at the invitation of Prof. Khabat Heshami).

Summer and Winter Schools

2023 Compute Ontario Summer School (COSS) Toronto, Canada
Summer School Student Jun 2023

- Attended courses on high-performance computing and scientific applications using Python and Julia, solving assignments with an average grade of 98%. Gained certifications in machine learning, artificial neural networks, ARC (Advanced Research Computing) in Julia, and HPC (High Performance Computing) in Python.

2025 Winter School on Quantum Networks Arizona, USA
Winter School Student Jan 2025

- Attended courses on the theory of quantum channels, principles of quantum networks, integrated photonic platforms for quantum networks, and optical communication channels.